CHAPTER 7

Absorption of Nuclear Radiation

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Our understanding of the nature of nuclear particles is based on their mode of interaction with matter. If the interaction of the various nuclear particles with various material is known it is possible to use this knowledge for e.g. proper utilization and construction of detection and measuring devices for radiation, the design of radiation shielding, the medical and biological applications of radiation, radiochemical synthesis, etc.

The term *nuclear radiation* is used to include all elementary particles, both uncharged (e.g. photons) and charged (e.g. electrons), having energies in excess of approximately 100 eV. This is valid whether the particles have been produced through nuclear reactions (spontaneous or induced) or have acquired their energy in electrostatic accelerators.

Radiochemistry and Nuclear Chemistry
ISBN 978-0-12-405897-2, http://dx.doi.org/10.1016/B978-0-12-405897-2.00007-0
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100 eV is a high energy in comparison to ionization energies (usually <15 eV) and to the energies involved in chemical bonds (normally 1−5 eV). Therefore, nuclear radiation can cause several ionizations in its passage through matter; this is reflected in the common name *ionizing radiation*. Neutrons of energies <100 eV are included because their absorption (capture) by nuclei results in emission of nuclear radiation with energies much greater than 100 eV.

The passage of such high energy radiation through matter results in the transfer of energy to the atoms and molecules of the absorber material. This transfer of energy continues until the impinging particle of the radiation has reached the same average kinetic energy as the atoms comprising the material; i.e. until *thermal equilibrium* is obtained.

In considering the absorption of nuclear radiation it is appropriate to view the overall process from two aspects: (1) from the impinging particles point of view, i.e.processes occurring to the nuclear particles themselves as their energies are reduced to the thermal equilibrium value; such *absorption processes* are the principal consideration of this chapter; (2) from the absorbing materials point of view, i.e. processes in the absorbing material due to the effect of the transfer of energy. This transfer results initially in excitation and ionization which cause physical and chemical changes. The study of these effects is the domain of *radiation chemistry* and is considered in Chapter 8.

7.1. SURVEY OF ABSORPTION PROCESSES

The reduction in the intensity of a beam of ionizing particles can be caused either by reaction with the nuclei of the absorbing material (nuclear reactions) or with the atomic electrons (electron collision). In Table 7.1 the most important processes involved in the absorption of nuclear radiation in matter are listed along with the probability for each process. Comparison shows that the probability of interactions with electrons is considerably greater than that of a nuclear reaction; the only exception to this is the case of neutron absorption. In fact the principal mode of interaction between the particle and the atoms of the absorbing material involves the electromagnetic fields of the particle and the atomic electrons. Since neutrons are neutral particles which do not with electrons, in order for them to transfer energy to an atom it is necessary that they experience a direct collision with a nucleus. Consequently for all particles except neutrons, nuclear reactions can be neglected in considering the processes involved in the reduction of the intensity of the particle beam.

As nuclear radiation passes the atoms of an absorber, it can transfer some of its energy to the atomic electrons. If the amount of energy transferred is sufficient, ionization of the atom results. The positive ion and the electron thus formed are known as an *ion pair*. Frequently the electrons from this primary ionization have sufficiently high kinetic energy to cause *secondary ionization* in other atoms. The number of electrons produced in